NTU PaCE x Cybrdeck

Applied Industry Project Briefs (DRAFT)

Programme Track: SkillsFuture Career Transition Programme (SCTP) — Data Science & AI (Learner Sprints)
Industry Partner: Cybrdeck Systems
Engagement Format: 6-Week Applied Sprint (Three 2-Week Iterative Cycles)
Team Structure: Recommended Squads of 2–3 Builders
Mentorship Cadence: Weekly 90-minute technical standup / code review plus asynchronous repository feedback
Lead Mentors: Gwendalynn Lim (Tech Lead / Quantum Systems Architect), Benjamin Lim (Co-Founder)

Brief 1: Schema Parsing Pipeline for Enterprise Workflow Automation

Track: Ingestion & Data Engineering | **Client Case:** Cybrdeck Workflow Systems

1. Executive Context & Problem Statement

Cybrdeck operates workflow automation across heterogeneous, semi-structured inputs—intake forms, community registers, operational notes, and system event logs. Upstream data frequently arrives with missing keys, inconsistent structural nesting, and type ambiguity. Before downstream automation or analytical services can trigger, this data must pass through an automated ingestion boundary that guarantees data integrity and quarantines anomalies without breaking system availability.

2. Core Functional Deliverables

Learners design and deploy an automated ingestion pipeline that:

- **Ingests Raw Workflow Payloads:** Parses multi-format, semi-structured operational records.
- **Enforces Invariant Data Contracts:** Validates structural schemas, field types, and required attributes.
- **Executes Dead-Letter Quarantine:** Automatically catches malformed, missing, or corrupted records, routing them to an isolated quarantine with explainable diagnostic logs.
- **Exposes Structured Data Endpoints:** Delivers normalized, queryable outputs consumable by downstream reporting and automation engines.

3. Bi-Weekly Delivery Rhythm

- **Sprint 1 (Weeks 1–2) — Ingestion & Schema Contracts:** Establish baseline data contracts; implement automated parsing and validation logic; verify rejection handling for malformed test payloads.
- **Sprint 2 (Weeks 3–4) — Resiliency & Quarantine Routing:** Implement dead-letter triage and exception-handling logic; optimize ingestion throughput on larger batch volumes; generate automated validation logs.
- **Sprint 3 (Weeks 5–6) — Service Encapsulation & Showcase:** Package the pipeline into an isolated, deployable microservice; write architecture documentation; deliver an end-to-end live technical demonstration.

4. Technical Governance & Standards

- **Environment & Stack:** Open-standard scripting language, modern schema-validation libraries, and production-grade version control (tooling aligned with learner competency).
- **Provided Scaffolding:** Synthetic enterprise workflow datasets, schema test cases, and reference delivery criteria.
- **Definition of Done:** A working codebase with automated test verification, documented schema validation rules, an isolated runtime packaging setup, and a live demonstration parsing raw input into validated, queryable structures.

Brief 2: Anomaly Triage Pipeline for Enterprise Workflow Automation

Track: Operations & Decision Intelligence | **Client Case:** Cybrdeck Workflow Systems

1. Executive Context & Problem Statement

In live automated studio operations, exceptions inevitably occur—duplicate transactions, dropped webhook events, out-of-range system latencies, and stalled operational sequences. Without automated triage, teams face severe alert fatigue. Cybrdeck requires an intelligent triage engine that monitors workflow telemetry, detects operational anomalies, evaluates severity, and routes incidents to appropriate resolution tiers.

2. Core Functional Deliverables

Learners design and deploy an operational triage engine that:

- **Monitors Telemetry Streams:** Ingests continuous event logs and transaction records from automated workflows.
- **Detects Deviations & Faults:** Applies statistical and heuristic detectors to identify operational anomalies against normal system baselines.
- **Automates Severity Scoring:** Prioritizes incidents using multi-factor scoring (business impact, frequency, operational risk).

- **Routes Triage Actions:** Categorizes incidents into deterministic outcomes (Auto-Resolve, Human Review, Immediate Escalation) with a clear audit trail.

3. Bi-Weekly Delivery Rhythm

- **Sprint 1 (Weeks 1–2) — Baseline Modeling & Signal Extraction:** Ingest synthetic telemetry; analyze normal operating bounds; implement baseline rule-based and statistical anomaly detectors.
- **Sprint 2 (Weeks 3–4) — Triage Scoring & Routing Matrix:** Build severity-scoring algorithms; implement routing rules to filter false-positive noise; generate structured incident payloads.
- **Sprint 3 (Weeks 5–6) — Service Encapsulation & Showcase:** Wrap triage logic into an isolated, deployable service; output structured audit trails; deliver an end-to-end incident handling demonstration.

4. Technical Governance & Standards

- **Environment & Stack:** Open-standard numerical/statistical libraries, modern web service frameworks, and production-grade version control (tooling aligned with learner competency).
- **Provided Scaffolding:** Synthetic telemetry and transaction event logs with seeded fault scenarios, plus reference architectural rubrics.
- **Definition of Done:** A functional triage pipeline that reliably isolates seeded operational faults, outputs clear human-readable diagnostic reasons, runs in an isolated environment, and is verified through a live incident response demonstration.

Brief 3: SME Legacy Database Modernization & Record Unification

Track: Data Engineering & Database Modernization | **Sector Case:** SME Digital Transformation

1. Executive Context & Problem Statement

Traditional small-to-medium enterprises (SMEs) frequently operate on legacy relational databases, exported accounting spreadsheets, and fragmented CRM archives accumulated over decades. These databases suffer from severe data degradation: duplicate client profiles, missing postal codes, unstandardized corporate entity names, and orphaned transaction logs. Before the enterprise can adopt modern CRM tools, automated customer outreach, or business intelligence dashboards, the legacy store must be audited, deduplicated, and transformed into a unified customer master record.

2. Core Functional Deliverables

Learners design and implement a data modernization and deduplication pipeline that:

- **Ingests Multi-Source Legacy Extracts:** Ingests legacy relational dumps (CSV/SQL/tabular records) representing historical customer accounts and transactions.
- **Executes Fuzzy Matching & Deduplication:** Identifies and clusters duplicate customer records across conflicting spellings, variations in business registration numbers, and phone/email discrepancies.
- **Performs Address & Profile Normalization:** Standardizes localization formats (e.g., postal codes, regional phone dialing codes, currency formatting).
- **Generates a Unified Customer Master Model:** Outputs clean, unified client entity tables with relationship links to past transaction histories, alongside an automated data-quality report quantifying deduplication gains.

3. Bi-Weekly Delivery Rhythm

- **Sprint 1 (Weeks 1–2) — Ingestion & Data Quality Profiling:** Load legacy database tables; profile data completeness, null-rates, and schema anomalies; design the target unified client schema.
- **Sprint 2 (Weeks 3–4) — Entity Resolution & Deduplication Logic:** Build rule-based and fuzzy string matching algorithms to group duplicate client profiles; establish deterministic conflict-resolution rules (e.g., newest timestamp vs. most complete record).
- **Sprint 3 (Weeks 5–6) — Target Schema Migration & Quality Audit:** Migrate cleaned records into the unified relational structure; generate automated before-and-after audit logs; deliver a technical walkthrough comparing legacy fragmentation against modern queryable records.

4. Technical Governance & Standards

- **Environment & Stack:** Open-standard relational database (SQL) and scripting language, modern data manipulation libraries, and version-controlled scripts (tooling aligned with learner competency).
- **Provided Scaffolding:** Fully synthetic, masked benchmark datasets exhibiting classic legacy data anomalies (duplicate entities, inconsistent schema keys, unnormalized fields). Zero proprietary or personally identifiable information (PII) is utilized.
- **Definition of Done:** A functional migration and deduplication pipeline that successfully ingests legacy data, resolves conflicting client entities, outputs a clean, unified relational database, and produces an auditable data-quality scorecard demonstrating measurable deduplication accuracy.